Experiment 19: Rate Determination and Activation Energy

An important part of the kinetic analysis of a chemical reaction is to determine the activation energy, E_a. Activation energy can be defined as the energy necessary to initiate a spontaneous chemical reaction. An example of activation energy is the combustion of paper. The reaction of cellulose and oxygen is spontaneous, but you need to initiate the combustion by adding activation energy from a lit match.

In this experiment you will investigate the reaction of crystal violet with sodium hydroxide. Crystal violet, in aqueous solution, is often used as an indicator in biochemical testing. The reaction of this organic molecule with sodium hydroxide can be simplified by abbreviating the chemical formula for crystal violet as CV.

$$CV^+ (aq) + OH^- (aq) \rightarrow CVOH (aq)$$

As the reaction proceeds, the violet-colored CV^+ reactant will slowly change to a colorless product, following the typical behavior of an indicator. You will measure the color change with a Vernier Spectrometer. You can assume that absorbance is directly proportional to the concentration of crystal violet according to Beer's law.

The molar concentration of the sodium hydroxide, NaOH, solution will be much greater than the concentration of crystal violet. This ensures that the reaction, which is first order with respect to crystal violet, will be first order *overall* (with respect to all reactants) throughout the experiment. This is because the concentration of hydroxide will not change much during the reaction, and therefore its concentration (and order) become part of the rate constant. The rate will only be dependent on the concentration of crystal violet. This is a kinetic trick known as "pseudo first-order".

You will monitor the reaction at different temperatures, while keeping the initial concentrations of the reactants the same for each trial. In this way, you will observe and measure the effect of temperature change on the rate of the reaction. From this information you will be able to calculate the activation energy, E_a, or the reaction using the Arrhenius Equation.

OBJECTIVES

- React solutions of crystal violet and sodium hydroxide at four different temperatures.
- Measure and record the effect of temperature on the reaction rate and rate constant.
- Calculate the activation energy, E_a, for the reaction.

MATERIALS

LabQuest	0.10 M sodium hydroxide, NaOH, solution
1 L beaker	2.5×10^{-5} M crystal violet solution
Vernier Colorimeter or Spectrometer	ice
Temperature Probe or thermometer	two 10 mL graduated cylinders
50 mL beaker	two 100 mL beakers
5 plastic cuvettes	watch with a second hand

PROCEDURE

NOTE: You will be preparing a series of water baths during this experiment. Measure the temperature of the water baths as accurately as possible.

1. Obtain and wear goggles.

2. Prepare a *blank* by filling a cuvette 3/4 full with distilled water. To correctly use cuvettes, remember:
   - Wipe the outside of each cuvette with a lint-free tissue.
   - Handle cuvettes only by the top edge of the ribbed sides.
   - Dislodge any bubbles by gently tapping the cuvette on a hard surface.
   - Always position the cuvette so the light passes through the clear sides.

3. Connect the Spectrometer to LabQuest and choose New from the File menu.

4. Calibrate the Spectrometer.
   a. Place the blank cuvette in the Spectrometer.
   b. Choose Calibrate from the Sensors menu. Wait for the Spectrometer to warm up.
   c. Select Finish Calibration. When the message "Calibration completed" appears, select OK.
   d. It is probably a good idea to tap the graph tab and tap the "play" button to see a very flat full spectrum for your blank water. Then tap stop. Your spectrophotometer is now calibrated and you are now ready to conduct experiments. Do not tap "file" and "new", otherwise lab quest will think your spectrophotometer needs to get calibrated again.

5. Determine the optimum wavelength for examining the crystal violet solution and set up the data-collection mode.
   a. Empty the blank cuvette and rinse it twice with small amounts of 2.0×10^{-5} M crystal violet solution. Fill the cuvette about 3/4 full with the crystal violet solution and place it in the spectrometer.
   b. Start data collection. A full spectrum graph of the solution will be displayed. Stop data collection. Move the cursor over he wavelength of maximum absorbance (λ max) to identify that wavelength.
   c. Tap the Meter tab. On the Meter screen, tap Mode. Change the mode to Time Based. The default data collection of 1 sample per second for 200 seconds will be suitable. Select OK.
   d. Check the wavelength indicated in the red box on the meter tab. Is it λ max? If it is not, tap the red box and tap "Change wavelength" to ensure you complete the experiment at λ max.
   e. Remove the cuvette from the Spectrometer and dispose of the crystal violet solution as directed. Save and use the same cuvette for Step 7. Proceed to Step 6.

6. Prepare a 25°C water bath in a 1 L beaker. The bath can be shallow because you will be using it to condition the two 10.0 mL samples of reactants.

7. Prepare the NaOH and crystal violet solutions for the first trial.
   a. Use a 10 mL graduated cylinder to obtain 10.0 mL of 0.10 M NaOH solution. Transfer the solution to a 100 mL beaker. **CAUTION:** *Sodium hydroxide solution is caustic. Avoid spilling it on your skin or clothing.*
   b. Use another 10 mL graduated cylinder to obtain 10.0 mL of 2.5×10^{-5} M crystal violet solution. Transfer the solution to a second 100 mL beaker. **CAUTION:** *Crystal violet is a biological stain. Avoid spilling it on your skin or clothing.*

c. Place the two 100 mL beakers of reactants in the water bath. Make sure that the level of the solutions on the beakers is below the level of the water bath. The beakers should be in the water bath for at least two minutes.

8. Conduct the first trial. Be ready to time the reaction.
   a. Check the temperature of the water bath; it should be holding steady at or near 25°C.
   b. When you are certain that the reactants are 25°C, pour the NaOH into the beaker of crystal violet solution. Start timing the reaction using the second hand on your watch. You will allow the reaction to run for one minute before proceeding to Part c of this step.
   c. After one minute, record the temperature of the reaction mixture.
   d. Fill a clean, dry cuvette about 3/4 full with the reaction mixture. Place the cuvette in the device (Colorimeter or Spectrometer). Close the lid on the Colorimeter.
   e. Start data collection. Observe the progress of the reaction. You may stop data collection early if desired.
   f. When data collection is complete, carefully remove the cuvette from the device. Dispose of the contents of the beaker and cuvette as directed.

9. Because the reaction is first order with respect to crystal violet, you can determine the rate constant, *k*, by plotting a graph of ln Absorbance *vs.* time.
   a. Tap the Table tab.
   b. Choose New Calculated Column from the Table menu.
   c. Enter the Name (ln Absorbance) and leave the Units field blank.
   d. Select the equation, "ALn(X)". Use Absorbance as the Column for X, and 1 as the value for A. Select OK. A graph of ln absorbance *vs.* time should now be displayed. Change the scale of the graph, if necessary.
   e. To calculate the best-fit line equation for your plotted data, choose Curve Fit from the Analyze menu. Choose Linear as the Fit Equation. The slope, *m*, is equal to the rate constant, *k*. Record this value in your data table. Select OK. You may need to highlight a region of the data (ideally within the first minute) to fit the linear equation to, before the absorbance value was too low to be accurate.

10. To prepare for the next trial, tap the File Cabinet icon to store the data from Trial 1.

11. Repeat the necessary steps to complete Trials 2–4. The next trials will be at ~20°C, ~15°C, and ~10°C. Use small amounts of ice to cool the water bath to the desired temperature.

12. Once you have completed all of the trials. Select "all runs" and print the graph of your four trials, either concentration vs time, or Ln vs time.

Experiment #19 Lab Report **Name:__________________**

Date:__________________ **Lab Partner: __________________**

DATA TABLE

Trial	Temperature (°C)	Temperature (K)	1/ T (1/K)	Rate constant, *k* (s^{-1})	Ln *k*
1	25.0			-0.0067194	
2	19.7			-0.0057167	
3	14.9			-0.0047806	
4	10.0			-0.0039483	

DATA ANALYSIS

1. Determine the activation energy, E_a, by plotting the natural log (Ln) of k *vs.* the reciprocal of absolute temperature. Fit a linear equation to this data. Determine what m and b are. Attach graph.

2. Calculate the activation energy, E_a, for the reaction. To do this, first calculate the best fit line equation for the data in Step 1. Use the slope, m, of the linear fit to calculate the activation energy, E_a, in units of kJ/mol. **Note:** On a plot of ln k *vs.* 1/absolute temperature, $E_a = -m \times R$.

4. A well-known approximation in chemistry states that the rate of a reaction often doubles for every 10°C increase in temperature. Use your data to test this rule. (**Note:** It is not necessarily equal to 2.00…..; this is just an approximate value, and depends on the activation energy for the reaction.)

5. Using the rate constant and precise temperature value for the trial that was done at room temperature (~20°C), as well as the E_a value you obtained in Step 3 above, calculate what the rate constant would be at 40°C.